



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

Department of Computer Science
Faculty of Computing

Project Task 5

(IP Addressing Scheme)

Programme	: Bachelor of Computer Science (<i>Data Engineering</i>)
Subject Code	: SECR1213
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Section : 02

Group : 3 – NotAvailable

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5.1 Get the Network Address from your lecturer.

Group/Section	Network Address
1	192.16.0.0/8
2	192.17.0.0/8
3	192.18.0.0/8
4	192.19.0.0/8
5	192.20.0.0/8
6	192.21.0.0/8
7	192.22.0.0/8
8	192.23.0.0/8
9	192.24.0.0/8
10	192.25.0.0/8

Figure 5.1.1

Figure 5.1.1 shows the list of network addresses of each group. The network address of our group is 192.18.0.0/16. We assume that our subnet mask is /16 or 255.254.0.0 to ensure that our range of address usage does not influence other group subnets.

Network Address details:

IP address (decimal)	192.18.0.0
IP address (binary)	11000000 00010010 00000000 00000000
Subnet Mask(decimal)	255.254.0.0
Subnet Mask(binary)	11111111 11111110 00000000 00000000

We partitioned the subnet mask into a network portion (16 bits) and a host portion (16 bits). By performing the AND operation of the IP address with the subnet mask, we identified the network address as 192.18.0.0 and the broadcast address as 192.18.255.255. The valid address range within our building spans from 192.18.0.1 to 192.18.255.254.

Network Address	192.18.0.0
Broadcast Address	192.18.255.255
Range of usable address	192.18.0.1 - 192.18.255.254

5.2 Divide it in the best possible way for your network.

5.2.1 Network distribution for different work area

Utilizing the classful addressing approach, we opted to allocate distinct subnets based on the defined work areas in task 4. Consequently, our building's network necessitates the creation of 5 subnets. To achieve this, an additional 3 bits ($2^3 = 8$) must be borrowed from the host portion for subnet construction.

Subnet	Network portion	Subnet bits	Host portion		IP address
0	11000000 00010010	000	00000 00000000	Network Address	192.18.0.0
			11111 11111111	Broadcast Address	192.18.31.255
1	11000000 00010010	001	00000 00000000	Network Address	192.18.32.0
			11111 11111111	Broadcast Address	192.18.63.255
2	11000000 00010010	010	00000 00000000	Network Address	192.18.64.0
			11111 11111111	Broadcast Address	192.18.95.255
3	11000000 00010010	011	00000 00000000	Network Address	192.18.96.0
			11111 11111111	Broadcast Address	192.18.127.255
4	11000000 00010010	100	00000 00000000	Network Address	192.18.128.0
			11111 11111111	Broadcast Address	192.18.159.255

Distribution of subnet for different work area:

Subnet	Work Area	Network Address	Broadcast Address	Range of usable address	Subnet Mask Address
0	Server Room & Meeting Room	192.18.0.0	192.18.31.255	192.18.0.0 - 192.18.31.255	/19 255.255.192.0
Ground Floor					
1	General Lab	192.18.32.0	192.18.63.255	192.18.32.0 - 192.18.63.255	/19 255.255.192.0
2	Cisco Network Lab & IOT Lab	192.18.64.0	192.18.95.255	192.18.64.0 - 192.18.95.255	/19 255.255.192.0
3	Student Lounge	192.18.96.0	192.18.127.255	192.18.96.0 - 192.18.127.255	/19 255.255.192.0
First Floor					
4	Office & Video Conferencing Room	192.18.128.0	192.18.159.255	192.18.128.0 - 192.18.159.255	/19 255.255.192.0

5.2.2 Network distribution for workstation in different work areas.

Ground Floor:

1. General Lab 1

Devices	IP Address
PC1	192.18.32.1
PC2	192.18.32.2
PC3	192.18.32.3
PC4	192.18.32.4
PC5	192.18.32.5
PC6	192.18.32.6
PC7	192.18.32.7
PC8	192.18.32.8
PC9	192.18.32.9
PC10	192.18.32.10
PC11	192.18.32.11
PC12	192.18.32.12
PC13	192.18.32.13
PC14	192.18.32.14
PC15	192.18.32.15
PC16	192.18.32.16
PC17	192.18.32.17
PC18	192.18.32.18
PC19	192.18.32.19
PC20	192.18.32.20
PC21	192.18.32.21

PC22	192.18.32.22
PC23	192.18.32.23
PC24	192.18.32.24
PC25	192.18.32.25
PC26	192.18.32.26
PC27	192.18.32.27
PC28	192.18.32.28
PC29	192.18.32.29
PC30	192.18.32.30
PC31	192.18.32.31
PC32	192.18.32.32
PJ1	192.18.32.33

2. General Lab 2:

Devices	IP Address
PC33	192.18.33.1
PC34	192.18.33.2
PC35	192.18.33.3
PC36	192.18.33.4
PC37	192.18.33.5
PC38	192.18.33.6
PC39	192.18.33.7
PC40	192.18.33.8
PC41	192.18.33.9
PC42	192.18.33.10
PC43	192.18.33.11

PC44	192.18.33.12
PC45	192.18.33.13
PC46	192.18.33.14
PC47	192.18.33.15
PC48	192.18.33.16
PC49	192.18.33.17
PC50	192.18.33.18
PC51	192.18.33.19
PC52	192.18.33.20
PC53	192.18.33.21
PC54	192.18.33.22
PC55	192.18.33.23
PC56	192.18.33.24
PC57	192.18.33.25
PC58	192.18.33.26
PC59	192.18.33.27
PC60	192.18.33.28
PC61	192.18.33.29
PC62	192.18.33.30
PC63	192.18.33.31
PC64	192.18.33.32
PJ2	192.18.33.33

3. Cisco Network Lab

Devices	IP Address
PC65	192.18.64.1
PC66	192.18.64.2
PC67	192.18.64.3
PC68	192.18.64.4
PC69	192.18.64.5
PC70	192.18.64.6
PC71	192.18.64.7
PC72	192.18.64.8
PC73	192.18.64.9
PC74	192.18.64.10
PC75	192.18.64.11
PC76	192.18.64.12
PC77	192.18.64.13
PC78	192.18.64.14
PC79	192.18.64.15
PC80	192.18.64.16
PC81	192.18.64.17
PC82	192.18.64.18
PC83	192.18.64.19
PC84	192.18.64.20
PC85	192.18.64.21
PC86	192.18.64.22
PC87	192.18.64.23
PC88	192.18.64.24

PC89	192.18.64.25
PC90	192.18.64.26
PC91	192.18.64.27
PC92	192.18.64.28
PC93	192.18.64.29
PC94	192.18.64.30
PC95	192.18.64.31
PC96	192.18.64.32
PJ3	192.18.64.33

4. IOT Lab

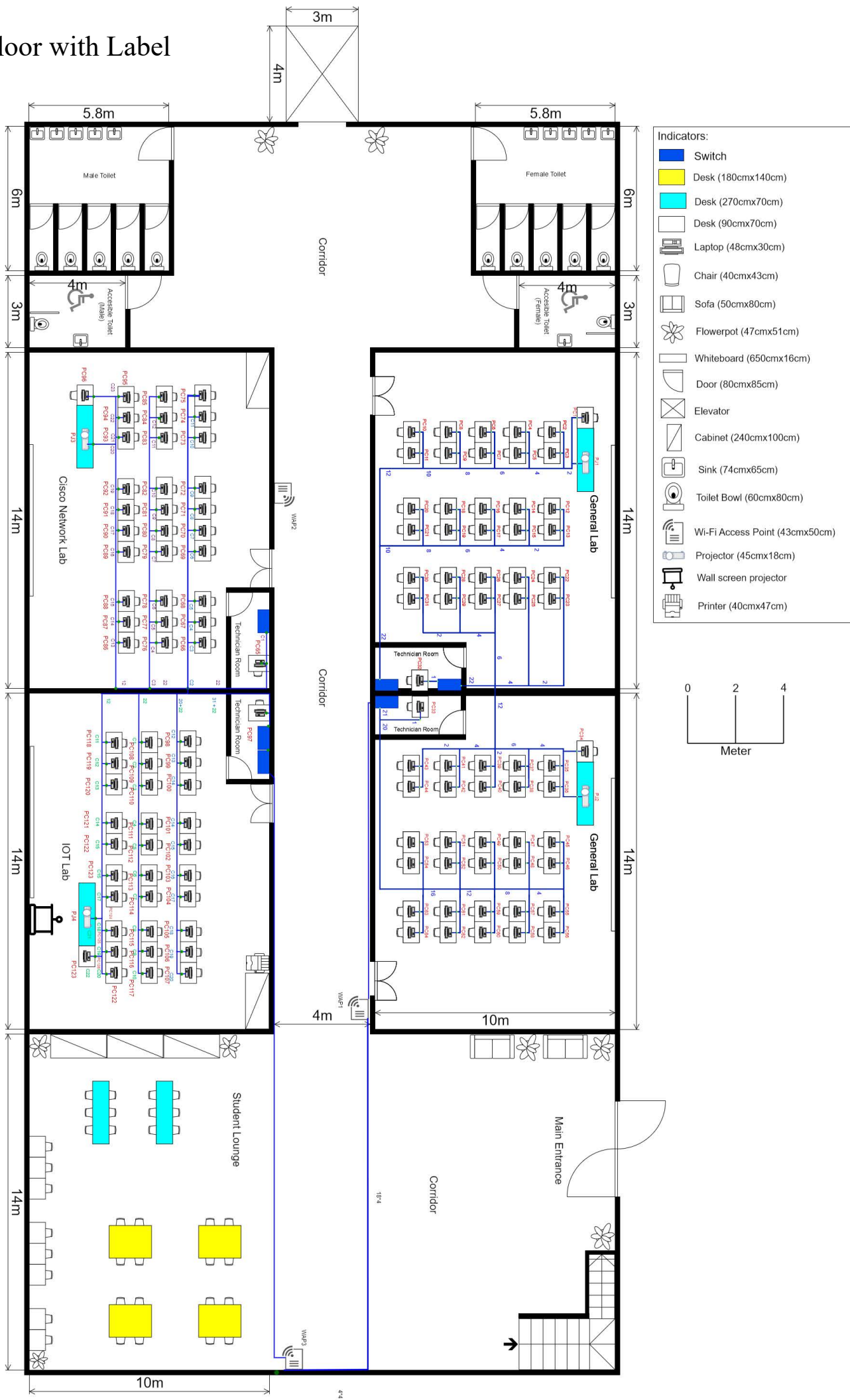
Devices	IP Address
PC97	192.18.65.1
PC98	192.18.65.2
PC99	192.18.65.3
PC100	192.18.65.4
PC101	192.18.65.5
PC102	192.18.65.6
PC103	192.18.65.7
PC104	192.18.65.8
PC105	192.18.65.9
PC106	192.18.65.10
PC107	192.18.65.11
PC108	192.18.65.12
PC109	192.18.65.13

PC110	192.18.65.14
PC111	192.18.65.15
PC112	192.18.65.16
PC113	192.18.65.17
PC114	192.18.65.18
PC115	192.18.65.19
PC116	192.18.65.20
PC117	192.18.65.21
PC118	192.18.65.22
PC119	192.18.65.23
PC120	192.18.65.24
PC121	192.18.65.25
PC122	192.18.65.26
PC123	192.18.65.27
PC124	192.18.65.28
PC125	192.18.65.29
PC126	192.18.65.30
PC127	192.18.65.31
PC128	192.18.65.32
PJ4	192.18.65.33

5. Student Lounge

Devices	IP Address
WAP3	192.18.96.1

Ground Floor with Label



First Floor:

1. Server Room:

Devices	IP Address
SV1	192.18.0.1
SV2	192.18.0.2
SV3	192.18.0.3
SV4	192.18.0.4
SV5	192.18.0.5
PC131	192.18.0.6

2. Staff Meeting Room

Devices	IP Address
PJ6	192.18.1.1

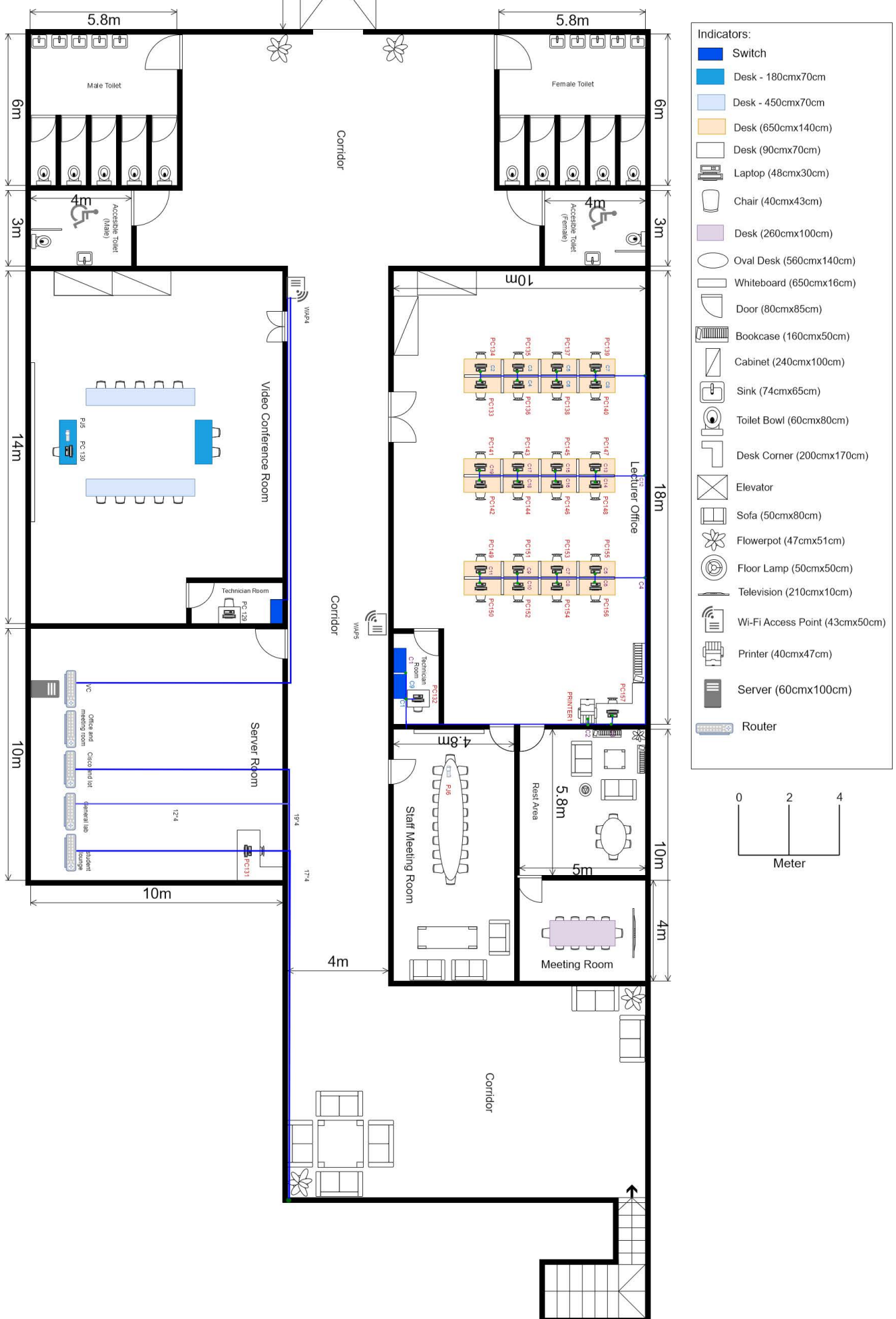
4. Video Conference Room

Devices	IP Address
PC129	192.18.128.1
PC130	192.18.128.2
PJ5	192.18.128.3
WAP4	192.18.128.4

3. Lecturer Office:

Devices	IP Address
PC132	192.18.129.1
PC133	192.18.129.2
PC134	192.18.129.3
PC135	192.18.129.4
PC136	192.18.129.5

PC137	192.18.129.6
PC138	192.18.129.7
PC139	192.18.129.8
PC140	192.18.129.9
PC141	192.18.129.10
PC142	192.18.129.11
PC143	192.18.129.12
PC144	192.18.129.13
PC145	192.18.129.14
PC146	192.18.129.15
PC147	192.18.129.16
PC148	192.18.129.17
PC149	192.18.129.18
PC150	192.18.129.19
PC151	192.18.129.20
PC152	192.18.129.21
PC153	192.18.129.22
PC154	192.18.129.23
PC155	192.18.129.24
PC156	192.18.129.25
PC157	192.18.129.26
PRINTER1	192.18.129.27



Meeting Minutes #1

Date/Time		26 DEC 2023 10.00 pm	
Location		Online (Google Meet)	
Agenda		1. Understand the details about task 5.	
		2. Discuss the way to divide the subnet.	
		3. Complete task 5 together.	
Meeting Host		Tan Jun Yuan	
Attendance			
Name		Time	Reason For Absence
Yong Wern Jie		2200	-
Chen Pyng Haw		2200	-
Koh Li Hui		2200	-
Tan Jun Yuan		2200	-
Minutes			
No.	Item Discussed	Details	Person-In-Charge
1	Discussion for task 5	- Jun Yuan shared the information about task 5 and its rubric.	Jun Yuan
		- All members read and investigate the task carefully.	
		- Solved the problem raised by group members together through discussion.	
2	Suggestion for subnetwork division	- All members suggest their prefer subnetwork division	

		- All members discuss and decide the one that best suits our design.	All members
3	IP Addressing Scheme Overview	- All members examine how IP addresses are currently assigned to each work area.	All members
4	Proceed to the parts of Task 5	- All members work on google docs together to complete task 5. - At the same time, if any of the group members question and identify any issues or problems then we find ways to encounter problems together.	All members
5	Meeting Ended	- At 0130, the meeting ended after all the discussion and work was done.	All members